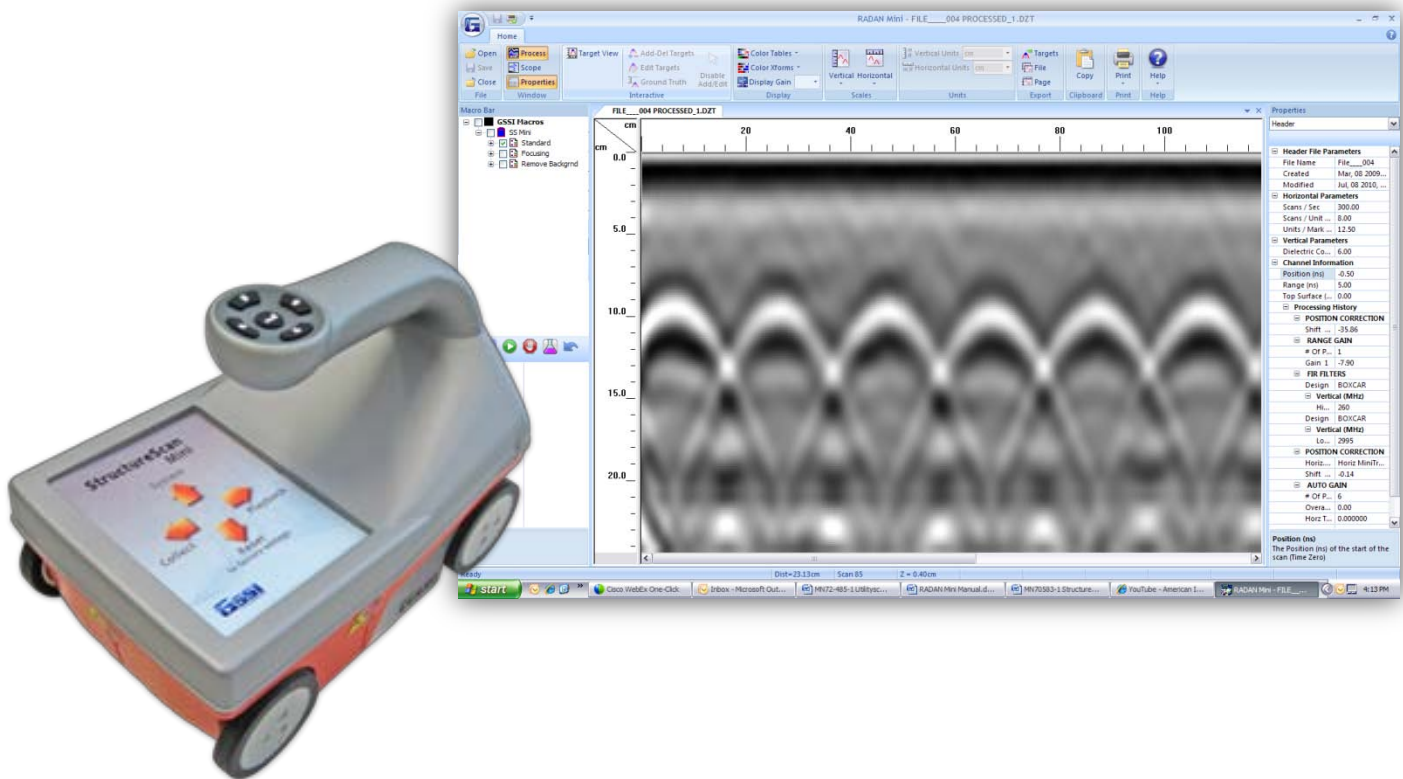




RADAN™ for StructureScan™ Mini

Geophysical Survey Systems, Inc.
www.geophysical.com

August, 2010



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Should you have any questions concerning this Agreement, you may contact GSSI by writing Geophysical Survey Systems, 12 Industrial Way, Salem, New Hampshire 03079 U.S.A.

You acknowledge that you have read this agreement, understand it and agree to be bound by its terms and conditions. You further agree that it is the complete and exclusive statement of the Agreement between you and GSSI which supersedes any proposal or prior Agreement, oral or written, and any other communications between us relating to the subject matter of this Agreement.

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Chapter 1: Getting Started

Thank you for purchasing RADAN™ for StructureScan™ Mini. The packing list included with your shipment lists all of the items in your order. The RADAN for StructureScan Mini program and example files are stored on a single CD-ROM disk.

System Requirements

- A Core i3 or better processor 2.0 GHz or greater recommended).
- GSSI will only support RADAN StructureScan Mini running on Windows 7, Vista or XP.
- 2 GB of RAM minimum (4 GB or greater recommended).
- A 256 MB video card running in 32-bit color mode that supports Open GL and has up-to-date video drivers. We recommend nVidia chipsets and Intel chipsets. Avoid any chipset from ATI.

GSSI Activation Policies

Your RADAN for StructureScan Mini license includes one license for installation on one computer.

General Description

RADAN for StructureScan Mini software was designed to process, view, and document data collected with the StructureScan Mini.

RADAN for StructureScan Mini module can perform the following functions:

- Copy images to third party software for documentation purposes
- Save images
- Print to all Windows supported printers
- Background Removal Filtering
- Ground Truth for near accurate depth calculation
- Dielectric calculation
- Colorize the data
- Add targets and export target information in an Excel format
- Gain (contrast)



How To Use This Manual

This manual is designed for both experienced and novice users of RADAN for StructureScan Mini. We recommend that all users read the entire manual.

Installing RADAN for StructureScan Mini

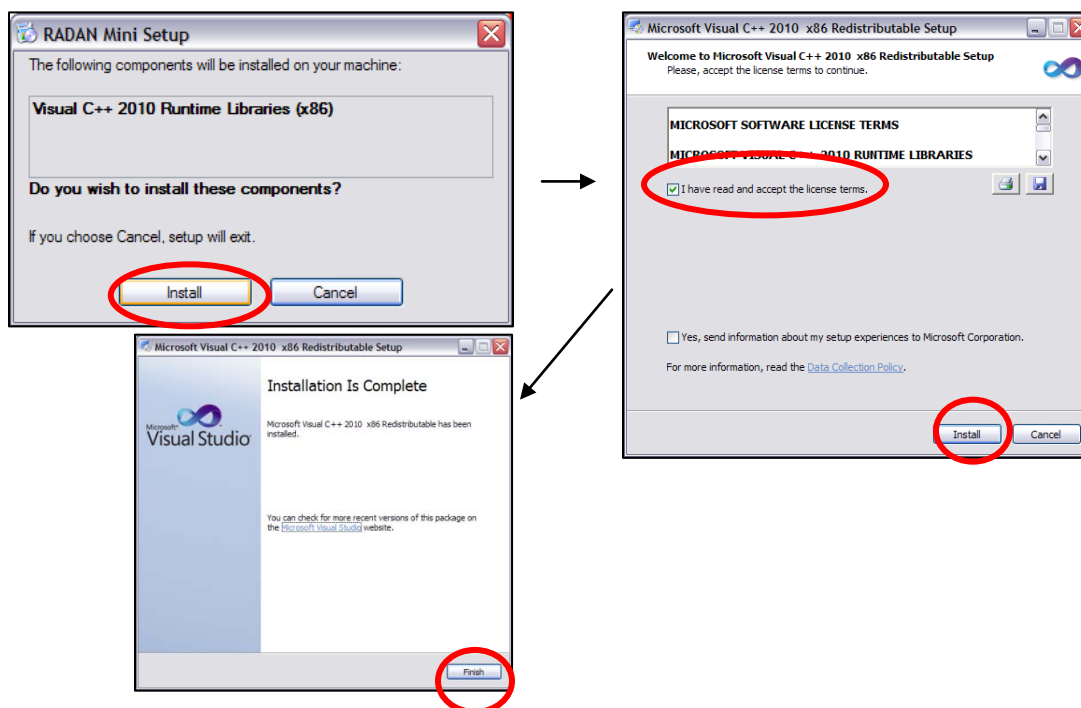
It is highly recommended that you are connected to the internet when installing the software so that activation can take place. If you are not, you will need to contact GSSI for other installation instructions.

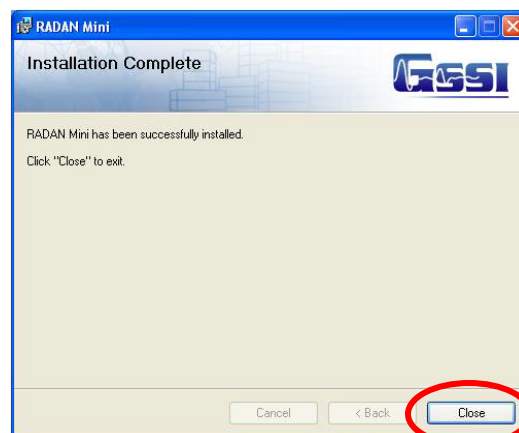
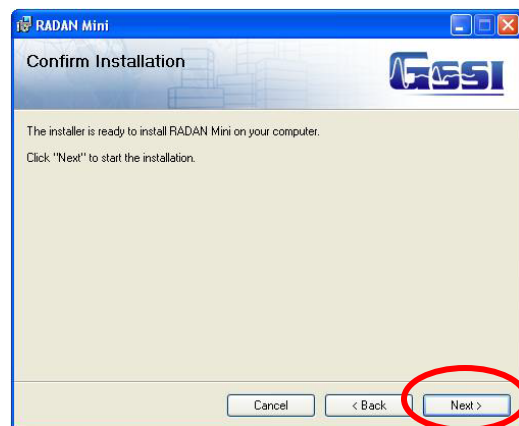
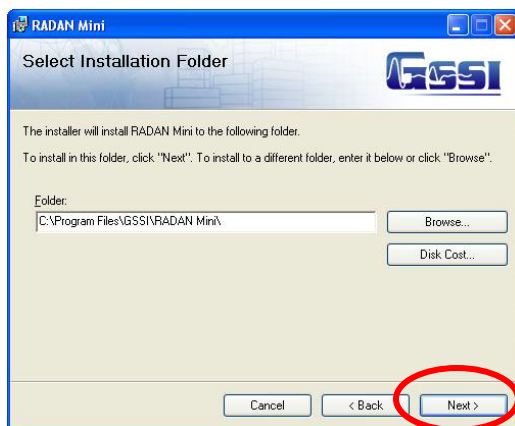
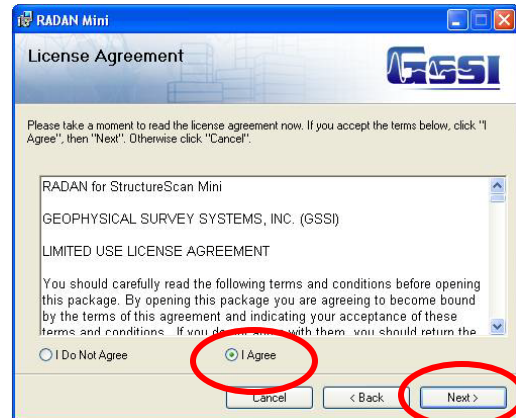
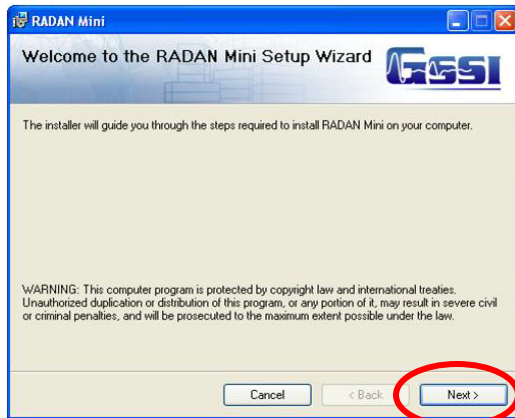
- 1** Insert the RADAN for StructureScan Mini Installation CD into your computer's CD drive.
- 2** The installation program should start automatically. Follow instructions on screen. If the program did start automatically, skip to step 5.
- 3** Using Windows Explorer, double-click the CD drive that contains the RADAN for StructureScan Mini Installation disk. It is probably Drive D: or E:.
- 4** Find the Start Icon and double-click it. 
- 5** Follow on-screen instructions:
 - **Install RADAN for StructureScan Mini:** Select this option to install the software onto your computer.
 - **Technical Support Web Site Info:** This will open a PDF file and has excellent information about our support web site and how to use this site. **It is highly recommended that you open, save and print this file.**
 - **GSSI Contacts:** This will open a PDF file on essential contacts for you here at GSSI. **It is highly recommended that you open, save and print this file.**



Installation Instructions

Click Install, Accept, Next, OK or Finish through all the screens. After you have selected "Install RADAN for StructureScan Mini" from the Main menu, these will be the screens you will see:





Important Note: Depending upon the age of your computer, your operating system (Windows XP, Windows Vista, or Windows 7), and the age of your operating system, you may get other screens asking you to install other software. You should accept these, or if you have questions, contact your IT person or GSSI.

Activating/Validating the RADAN for StructureScan Mini Software

Activating the Software

After installing the software, RADAN for StructureScan Mini will automatically start.

The first time you run the software, you will be asked to activate it. You **MUST** be connected to the internet to activate the software.

- 1** From the label located on the inside cover of the CD case, input the Product Key and the Serial Number into the appropriate fields.
- 2** The Computer Name is pre-filled by GSSI (this is the name of your computer from your computer system).
- 3** Enter a valid email address.
- 4** Click Activate/Validate your Product Key.



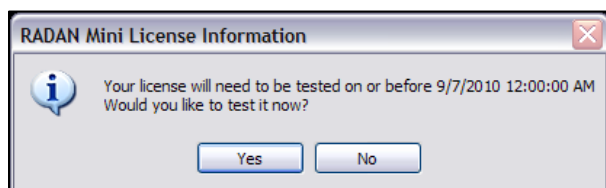
Note: If you choose not to activate your software at this time, the above screen will appear at the launching of the software until you activate the software.

Note: After activating the software, you do NOT have to be connected to the internet to run the software.

Validating the Software

At times, upon launching RADAN for StructureScan Mini, you will be asked to test the license. Click “Yes” and a pre-populated screen will appear. Simply click “Continue with RADAN Mini.”

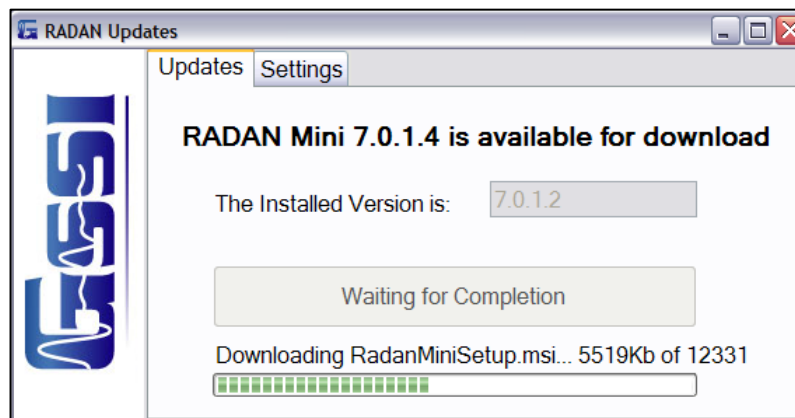
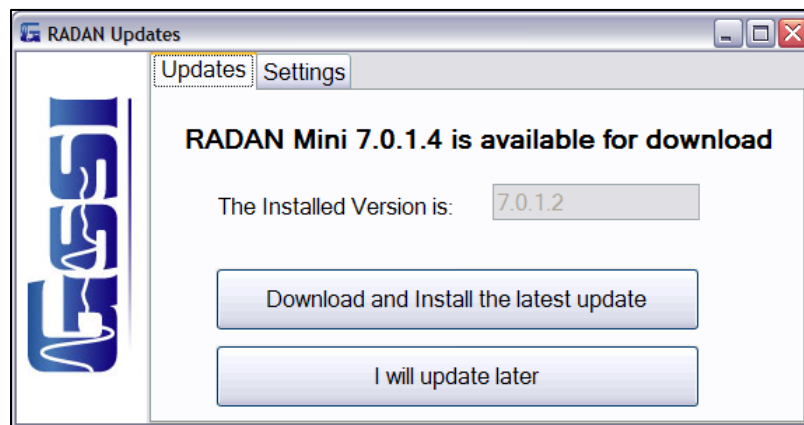
Note: You will need to be connected to the internet to complete this process.



Updating the Software

GSSI can automatically update the software if there is one available. Upon launching RADAN for StructureScan Mini AND if you are connected to the internet, you will be asked if you would like to load the update.

- Click “Download and Install the latest update” if you wish to update your software at this time.
- Click “I will update later” to update the software at a later date.



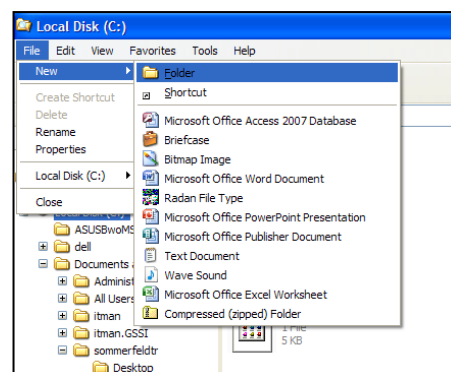
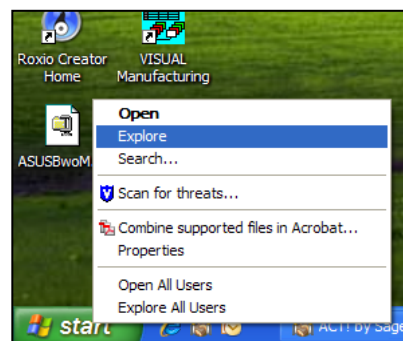
Chapter 2: Using RADAN for StructureScan Mini

Setting Up Your Laptop/PC and Data Transfer

Create folders on your PC/Laptop. A different folder should be created for each of your projects. Below is just a suggestion on how you might want to organize your folders. If you are comfortable with PCs and Windows, you can organize your folders however you wish.

Creating Folders on Your PC/Laptop

- 1** On your PC/Laptop, right-click **Start**.
- 2** Click **Explore**.
- 3** On the left side, click **Local Disk (C:)**.
- 4** Click **File** at the top of the screen.
- 5** Click **New**.
- 6** Click **Folder**.
- 7** Type in the name of your project and press Enter. This is where your source (original) data will be copied to.
- 8** Then create an output folder by:
 - a)** Double-clicking on the newly created folder to open it.
 - b)** Click **File**.
 - c)** Click **New**.
 - d)** Click **Folder**.
 - e)** Type "Output" and press Enter. This is where processed data will be located.
- 9** Close Explorer.



Transferring the Files

Copy the files you wish to view and/or process from the StructureScan Mini to the PC/Laptop.

- If you used the card to save data from the Mini, you may remove the card and insert the card into a card reader in your PC/Laptop. If you do not have a card reader, you will need to purchase one that will read the card from any technology store.
- If you did not use the card while collecting the data on your Mini, you may use ActiveSync from the Microsoft web site. This is free to download.

Regardless of the way you saved the data, you will need to copy the files from the StructureScan Mini to the Source folder you created previously.

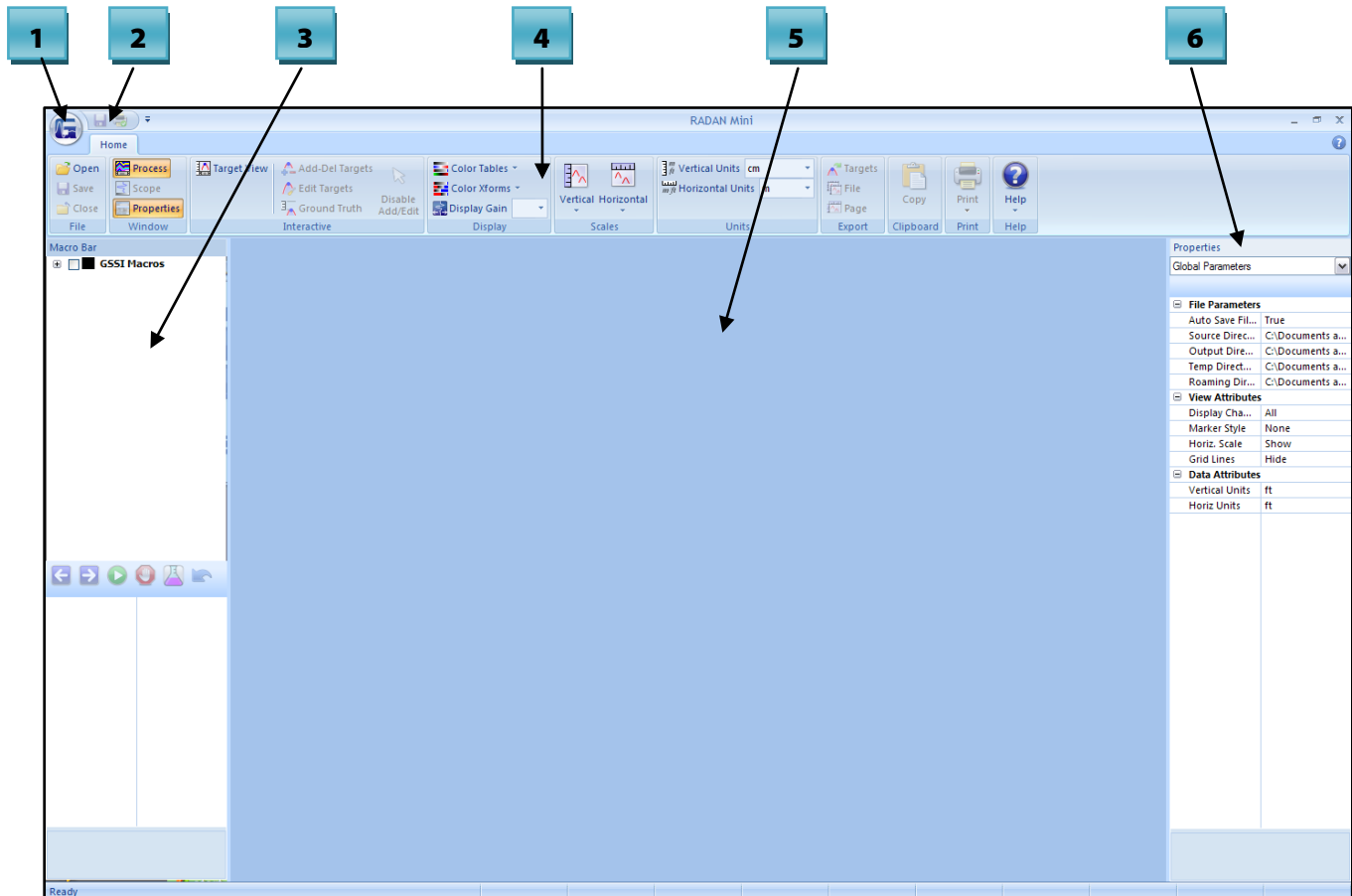
Launching and Navigating the Software

Launching the Software

Launch RADAN for StructureScan Mini by double-clicking the icon



The following screen is displayed:



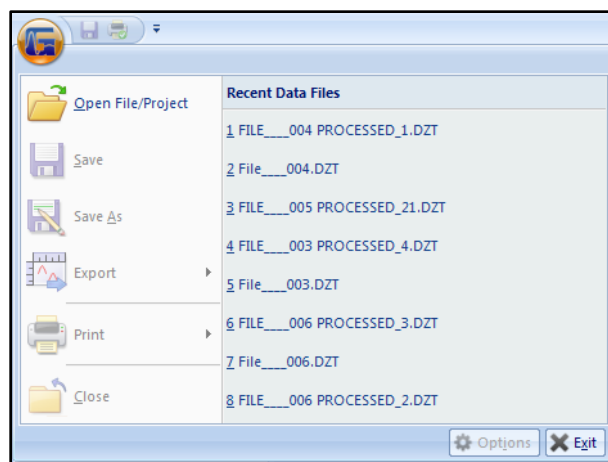
- 1** GSSI Button
- 2** Quick Access Toolbar
- 3** Macro Bar Screen
- 4** Ribbon
- 5** Data Screen
- 6** Properties Screen

Navigating Through the Main Screen

1. GSSI Button

Clicking on the button allows you to:

- Open a File/Project
- Save a File/Project
- Save As a File/Project under a different name
- Export your data
- Print your data
- Close a File/Project
- Open a previously process File/Project
- Exit RADAN for StructureScan Mini Software



2. Quick Access Toolbar

If you right-click any icon located in the GSSI button or in the Ribbon, you may add that icon to the Quick Access Toolbar. This is for options you use often.

3. Macro Bar Screen

The options located in this area will allow you to quickly apply to the data the more commonly used processes. A macro is a series of steps and/or options put together as one option. GSSI has created these commonly used processes as macros. This is detailed more in the Viewing and Processing the Data section.

4. Ribbon

This area contains all the options and processes you can do with your data. This is detailed more in the Viewing and Processing the Data section

5. Data Screen

This is where your data is displayed when you open a file.

6. Properties Screen

This area will display Global Parameters for all your files in a particular project, as well as Header information about a specific file (when opened).

Viewing and Processing the Data

Preparing the Software

Before opening a file, you will need to set up some Global Parameters located on the right side of the screen. Click on the following to change the parameters:

Auto Save File: Click here, then click on the down arrow to select one of the following values:

- **True:** Automatically save a processed file.
- **False:** User is prompted to save a processed file.

Source Directory: Click here, then click on the three dots to browse to the source folder where you copied your source (original) data.

Output Directory: Click here, then click on the three dots to browse to the output folder you created from your source folder.

Horiz. Scale:

- **Show:** Display horizontal scale.
- **Hide:** Do not display horizontal scale.

Grid Lines:

- **Show:** Display grid lines.
- **Hide:** Do not display grid lines.

Vertical Units: Units for the vertical scale.

Horiz Units: Units for the horizontal scale.

Note: Once you open a file, you cannot change any of the Global Parameters. You must close all files to make changes.

The screenshot shows the 'Properties' dialog box for 'Global Parameters'. It contains several sections: 'File Parameters', 'View Attributes', and 'Data Attributes'. Each section has a list of parameters and their current values.

Global Parameters	
File Parameters	
Auto Save Fil...	True
Source Direc...	C:\Documents a...
Output Dire...	C:\Documents a...
Temp Direct...	C:\Documents a...
Roaming Dir...	C:\Documents a...
View Attributes	
Display Cha...	All
Marker Style	None
Horiz. Scale	Show
Grid Lines	Hide
Data Attributes	
Vertical Units	ft
Horiz Units	ft

Opening a File

There are two ways to open a file.

- 1** Click the Open icon located on the left side of the Ribbon (Figure 1). This will display a screen defaulted to the folder you set in the Global Parameters->Source Directory.
 - Select a file to view/process and click Open (Figure 2),
 - Or double click on the Output Folder to open a previously processed file.

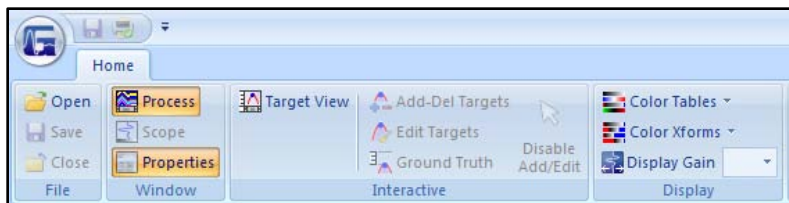


Figure 1

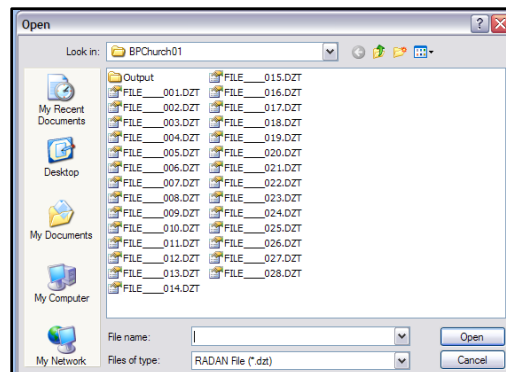


Figure 2

Or

- 2** Click the GSSI Button (Figure 3).
 - Here you may open any previously opened file,
 - Or click Open File/Project to select a file from your Source or Output folder (Figure 2).

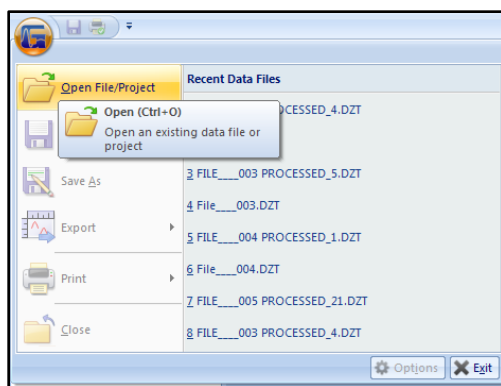


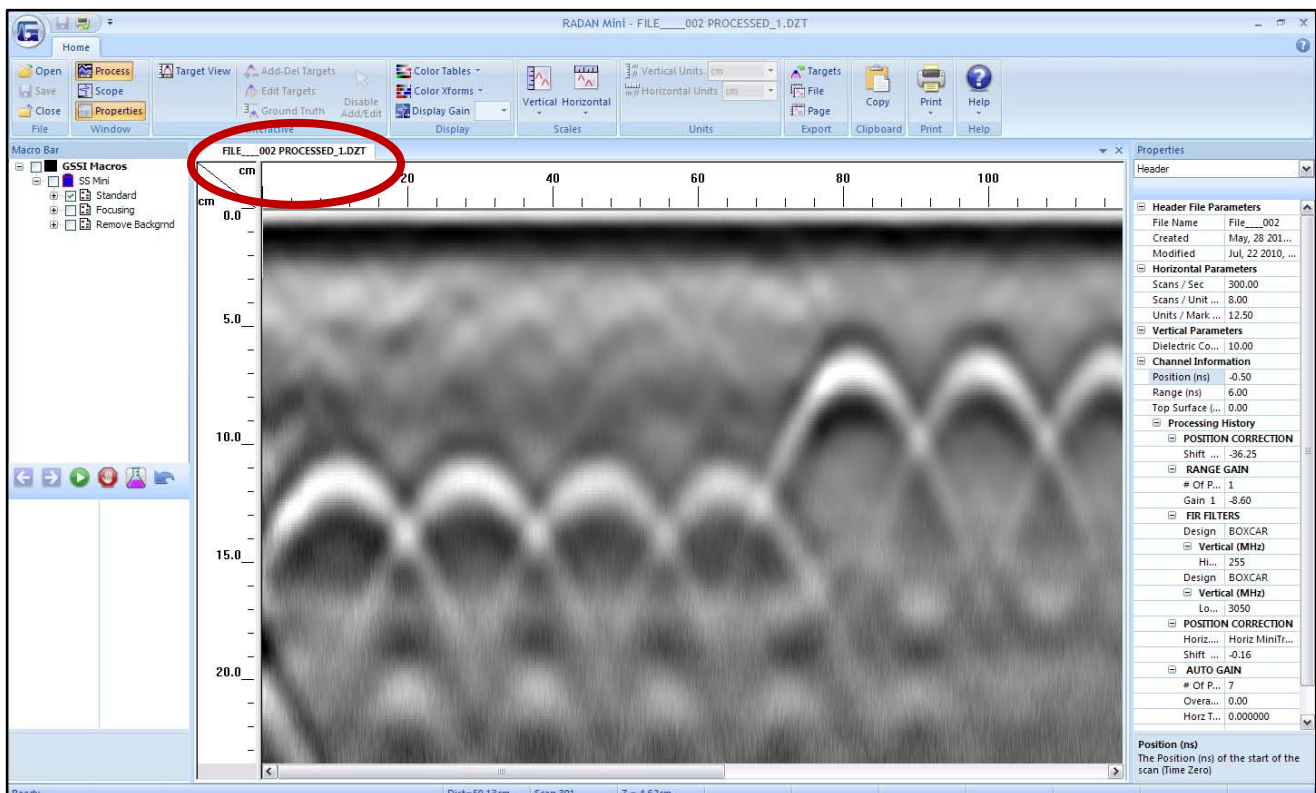
Figure 3

Processing the Data

Initial Process (automatically done by the software)

Once a file is opened, the software will automatically process the original, raw data and create a new processed file so that it is viewable. This process is called the Standard Macro located in the Macro Bar Screen.

- If you have configured Auto Save File set as **True**, the software will process the original file, save the processed file to the Output folder, and name the new file as “FileName”+”Processed”+”X”, where “X” represents the number of times the original file was opened and processed.
- If you have configured Auto Save file set as **False**, the software will process the file, however will ask you to save the processed file.
 - You may save the file using the software’s naming style,
 - Or you may rename the file to your style and save the file.
 - In either case, the processed file will be saved to the Output Folder.
- If the processed file was automatically saved or manually saved, **ONLY** the processed file is displayed.
- The original file still exists in the Source folder. This file remains untouched.
- Note the tab (red circle), this displays the name of the file being displayed.
- The Macro Bar screen displays the more commonly used processes which are called Macros.
- The Properties Screen now displays Header information, which is information about the file.



Ribbon Options

File

- **Open:** Open another file. This will create another tab for that processed file. Click on the tab for the file you want displayed. That will be the current tab.
- **Close:** Close the file located in the current tab.

Window

- **Process:** Toggles with Target View, Process allows you to select macros in the Macro Bar Screen to process the data.
- **Scope:** Toggle on and off to display the “O-Scope” RADAR signal (Figure 4). You may hold the left mouse button down on the data and move the mouse to view the RADAR signal at that position.
- **Properties:** Toggle to display the Properties screen.

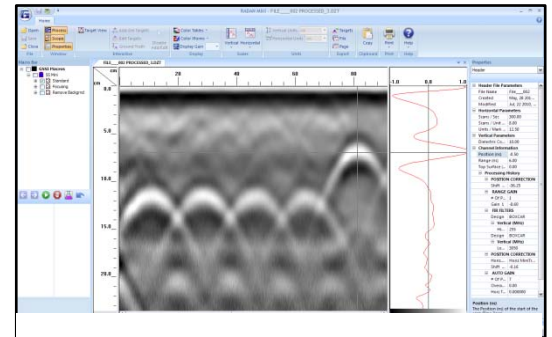


Figure 4

Interactive

- **Target View:** Toggle with the Process icon. Target view will allow the user to insert dots and ground truth information with the targets.
- **Add/Del Targets:** This will allow users to “tag” targets with a dot (Figure 5). Simply turn this on and click on a target to insert a dot. Right click on a dot to delete the dot. These dots will contain information about the target such as a horizontal distance, depth, and velocity. The information can be exported as a comma delimited ASCII file.
- **Edit Targets:** This will allow the user to click on a dot to change the color and size of the dot.
- **Ground Truth:** This will allow the user to click on a target and enter the KNOWN DEPTH of that target. This will adjust the scale accordingly.
- **Disable Add/Edit:** This will turn off Add/Del Targets, Edit Targets, or Ground Truth, whichever is on at the time so that you can use the crosshairs.

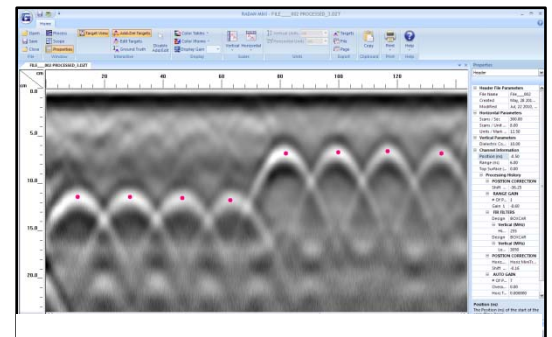


Figure 5

Display

- **Color Tables:** Change the color of your data.
- **Color Xform:** Change how the colors of your chosen color table is distributed.
- **Display Gain:** Change the contrast (brightness) of your data. This changes the display only, it does not change the actual data.

Scales

- **Vertical:** Change the vertical scale to Samples, Time (ns), or Depth.
- **Horizontal:** Change the horizontal scale to Distance, Scans, or None.

Units: Only available when NO files are open (or if you reopen RADAN and select an already processed file).

- **Vertical:** Change the vertical scale units.
- **Horizontal:** Change the horizontal scale units.

Export

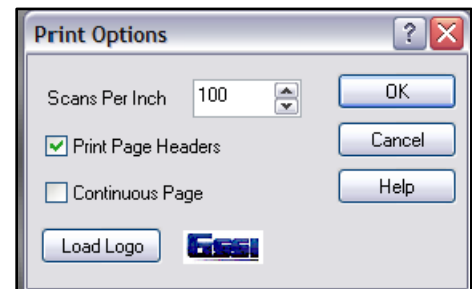
- **Targets:** Export targets (dots) information (distance, depth and layer velocity) to a comma delimited ASCII file. The name of the file is “FileName”+”X”.csv located in the output folder.
- **File:** Save the current file as a JPEG file (Picture File). This file will be named “FileName”+”X”.jpg located in the output folder.
- **Page:** Save the current page (what is displayed) as a JPEG file (Picture File). This file will be named “FileName”+”X”.jpg located in the output folder.

Clipboard

- **Copy:** Click this to copy the current file to the clipboard. This image then can be “pasted” to any other 3rd party software (PowerPoint, Word, etc.).

Print

- **Quick Print:** Print the image as is.
- **Setup:** Configure the printout.
 - Scan Per Inch: Number of scans per inch
 - Print Page Headers: Print Header information
 - Continuous Page: If data takes more than one page, print data on multiple pages or one page
 - Load Logo: Load and print your own logo, or GSSI logo
- **Print Preview:** Preview the printout on screen before actually printing the data.



Help

- **About:** Display the current version of the software.
- **Update:** Update the software to the latest version. You MUST be connected to the internet to do this.

Macro Bar Screen

A macro is a series of steps and/or options put together as one option. GSSI has created these commonly used processes as macros.

With the Process Icon turned on in the Ribbon, you may select the following macros:

Standard: This is the process that was done automatically when the file was opened. This process was done automatically by GSSI upon opening a file. You do not need to re-do this process.

Focusing: This macro will assist you in determining the dielectric and velocity of the RADAR signal for a closer accuracy of depth.

- 1** Move the bar located at the bottom of the Macro Bar screen to the left to increase the dielectric and decrease the velocity, or to the right to decrease the dielectric and increase the velocity.
- 2** Click the Test icon to see what the data will look like. Your goal is to turn the targets you see into oblong shaped dots. If you do not see oblong dots, click the Reset icon and repeat the process. You can click on the Stop icon to end the test.
- 3** Once you see the oblong dots, click the Go icon. This will save the data as a new processed file. If you have Automatically Save File configured to True, it will save the file with the "FileNameX"+"y", where "y" is the number of times you processed this file. If you configured Automatically Save File set as false, you will be asked to save the file with the system name or a name you select.
- 4** A new tab is created with this new processed file.



Go Icon



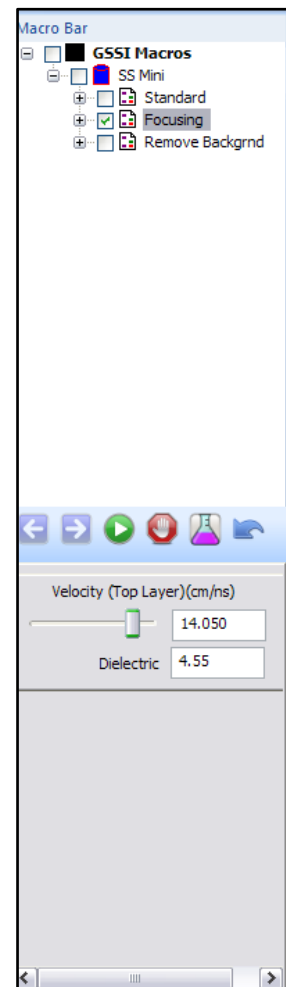
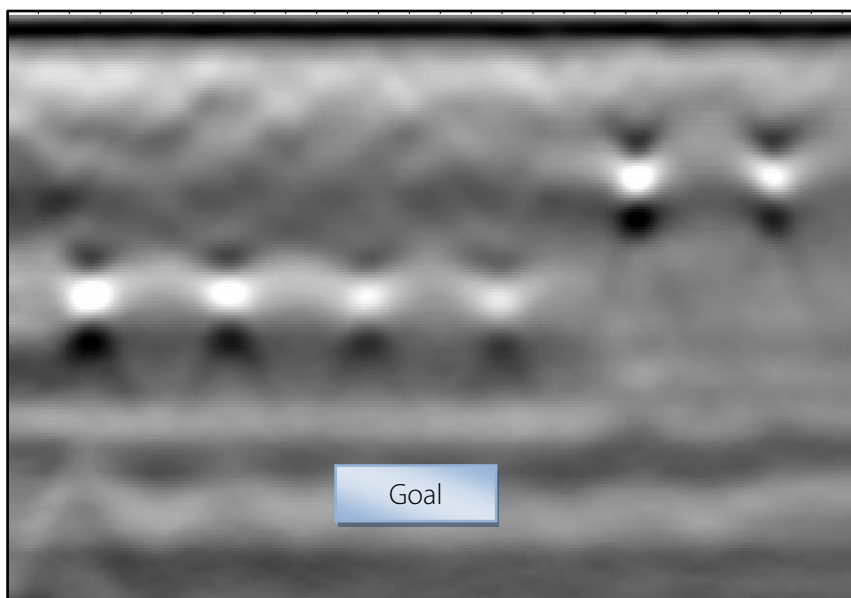
Stop Icon



Test Icon



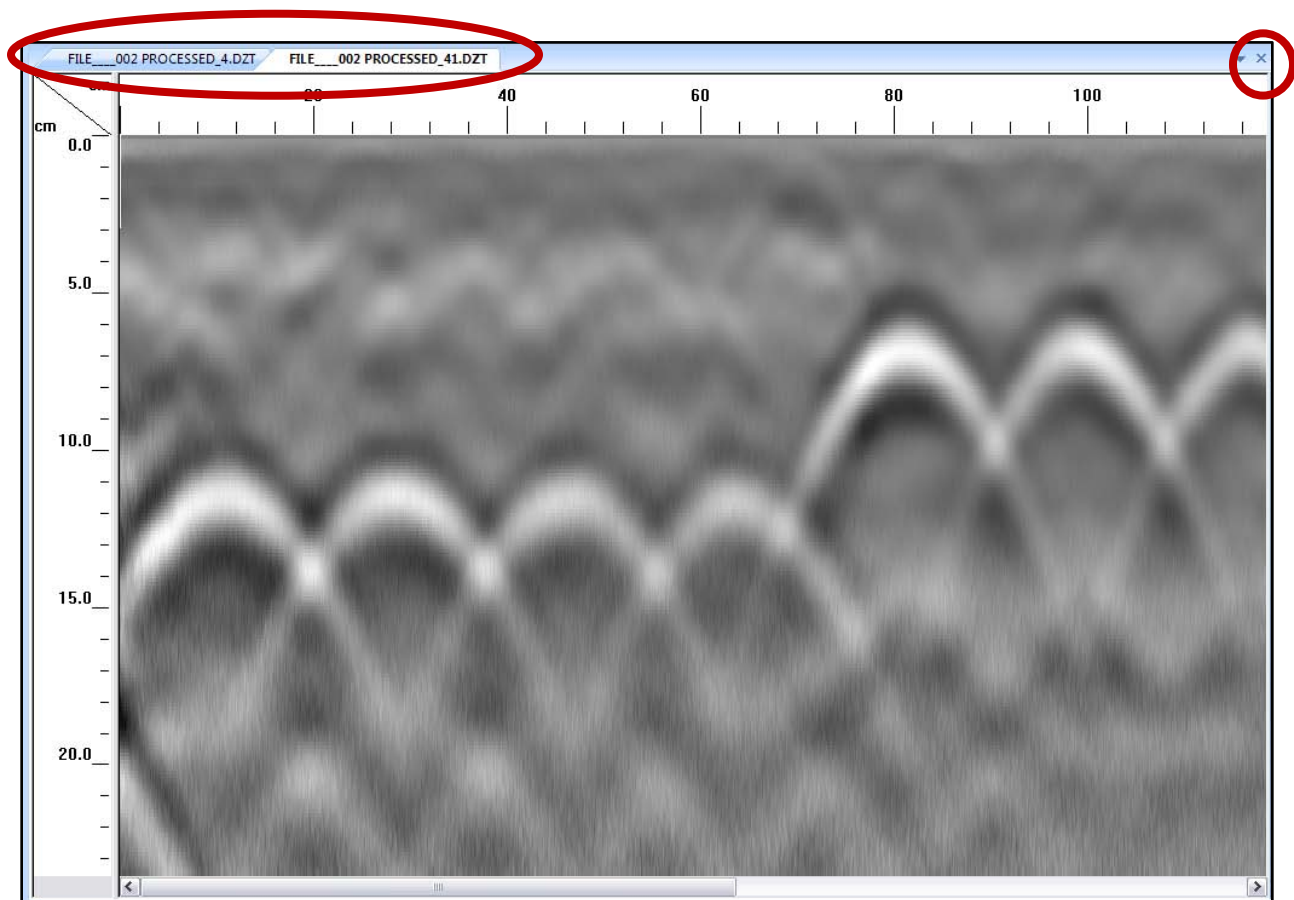
Reset Icon



Remove Backgrnd: This will remove any background noise that may be in your data. Usually this is horizontal banding. Unless you have this noise and CANNOT see your targets, you do not have to do this process.

- Click the Test icon to see what the data will look like. See if there is any difference in this data and the previous data by clicking on the Test and Reset icon a few times.
- If there is a difference, then click the Go icon. This will save the data as a new processed file. If you have Automatically Save File configured to True, it will save the file with the "FileNameX"+"y", where "y" is the number of times you processed this file. If you configured Automatically Save File set as false, you will be asked to save the file with the system name or a name you select.
- A new tab is created with this new processed file.

Here is a screen with an example displaying more than one tab. Click on a tab to display that data. You may also click on the "x" to close that current tab.



More the One Tab Example

Chapter 3: Tutorials

Saving StructureScan Mini Files as Pictures for a Report

RADAN for StructureScan Mini allows you to open a file, and save that file as a jpeg file (picture file). This will enable you to use that picture in any documentation (report, brochures, web sites, etc)

- 1** Open a file.
- 2** Optional steps (depending on what you would like to use for your documentation, and you may save more than one jpeg file):

a) Ribbon Options

- **Process:** Toggles with Target View, Process allows you to select macros in the Macro Bar Screen to process the data.
- **Scope:** Toggle on and off to display the “O-Scope” RADAR signal. (You may hold the left mouse button down on the data and move the mouse to view the RADAR signal at that position.
- **Target View:** Toggle with the Process icon. Target view will allow the user to insert dots and ground truth information with the targets.
- **Add/Del Targets:** This will allow users to “tag” targets with a dot. Simply turn this on and click on a target to insert a dot. Right click on a dot to delete the dot. These dots will contain information about the target such a horizontal distance, depth, and velocity. The information can be exported as a comma delimited ASCII file.
- **Edit Targets:** This will allow the user to click on a dot to change the color and size of the dot.
- **Ground Truth:** This will allow the user to click on a target and enter the KNOWN DEPTH of that target. This will adjust the scale accordingly.
- **Color Tables:** Change the color of your data.
- **Color Xform:** Change how the colors of your chosen color table is distributed.
- **Display Gain:** Change the contrast (brightness) of your data. This changes the display only, it does not change the actual data.

b) Macro Bar Screen

- **Focusing:** This macro will assist you in determining the dielectric and velocity of the RADAR signal for a closer accuracy of depth.
- **Remove Backgrnd:** This will remove any background noise that may be in your data. Usually this is horizontal banding. Unless you have this noise and CANNOT see you targets, you do not have to do this process.

- 3** To save the file displayed as a jpeg file:
 - a)** In the Export section of the Ribbon, click File.
 - b)** Click Save. This will save the file with a .jpg extension in the output folder.

Adding Targets and Output Targets as a CSV

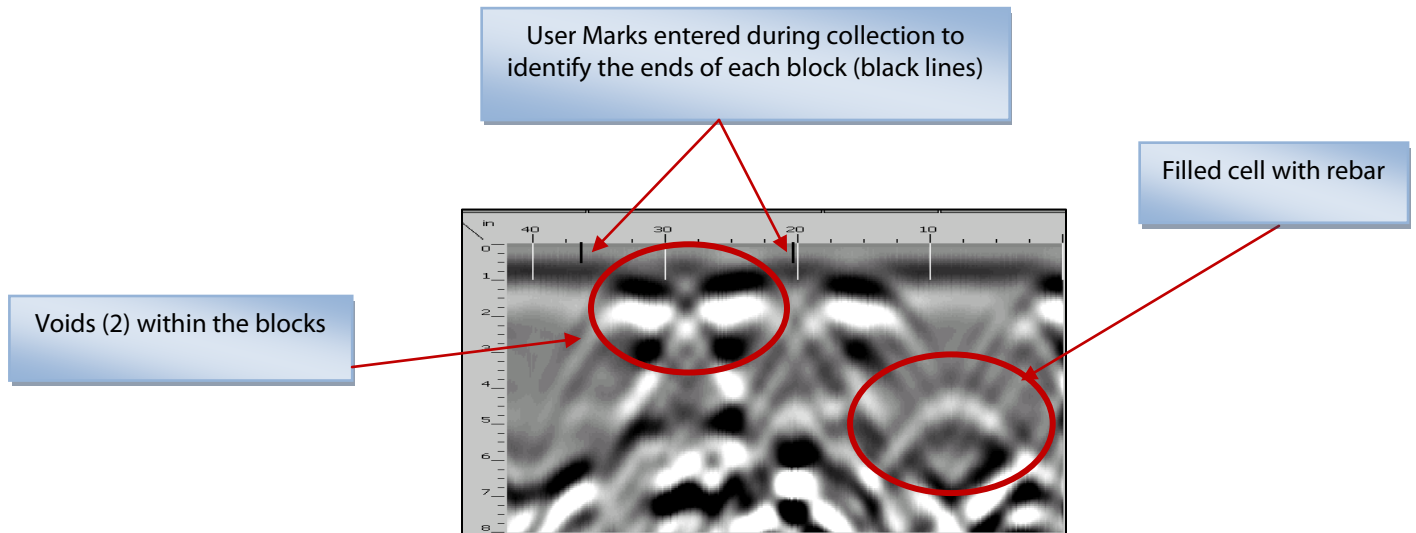
You may also add “dots” to your targets and export that information (distance, depth, and layer velocity) to a comma delimited file.

- 1** Open a file.
- 2** In the Window section of the Ribbon, click Target View.
- 3** Click Add-Del Targets.
 - a)** Click on a target to add a “dot.”
 - b)** Right-click on a “dot” to delete it.
- 4** Click Edit Targets.
 - a)** Click on a “dot” to change the color and size of the dot.
- 5** Click Ground Truth (If you have a known measurement for depth).
 - a)** Click on a target of that known measurement.
 - b)** Enter the known depth.
- 6** In the Export section of the Ribbon:
 - a)** Click on Targets.
 - b)** Save the file. It will be saved with a CSV extension located in your output folder.

Chapter 4: Sample Data

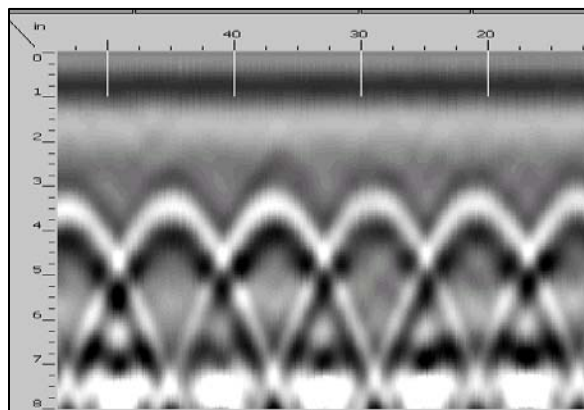
These examples are presented for informational use only. Conditions may be different at your site that may cause the images to look different from the data shown here.

Example 1 – Concrete Block Data



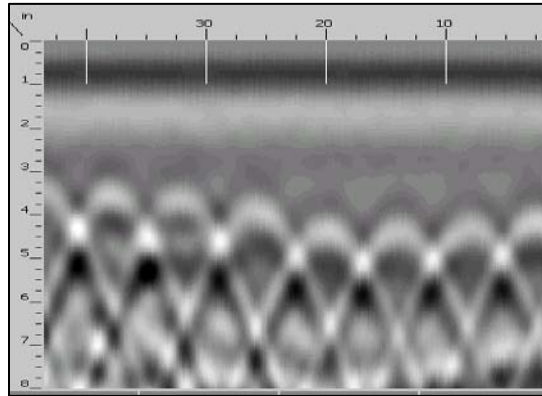
Example 2 – Rebar Data

This is an example of rebar. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 8 inches. Also note the approximate depth, the brightness of the hyperbolas, and they tend to be at a constant depth over the short term.



Example 3 – Mesh Data

This is an example of mesh. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 6 inches. Also note the approximate depth, the brightness of the hyperbolas, and mesh may be more uneven than rebar.



Example 4 – PVC

This is an example of two PVC pipes. Note they tend to be dimmer than rebar.

